THE TRIAD OF HOUSEHOLD WELL-BEING

Why Maternal Health, Sanitation, and Clean Cooking Must Be Governed as an Interlocking Public Health System

EXECUTIVE SUMMARY FOR DECISION-MAKERS

To evaluate India's readiness for the 2030 Sustainable Development Goals, this brief examines three foundational household well-being indicators: Maternal Mortality Ratio (SDG 3.1), Safely Managed Sanitation (SDG 6.2), and Clean Cooking Access (SDG 7.1). Using out-of-sample time-series validation (2000–2024), our models reveal a critical governance reality: while clean cooking adoption approaches universal targets (99.0% forecast), Safely Managed Sanitation plateaus at 76.2% (missing the 100% benchmark), and Maternal Mortality displays profound volatility (point forecast 94.3 against a <70 target, with a 95% prediction interval spanning 7.0 to 181.7). These three variables do not operate in silos; they form an interlocking domestic health ecosystem where household pathogen exposure and time poverty directly dictate maternal survival. Government intervention must pivot from siloed infrastructure schemes to integrated, risk-aware delivery.

1. Why These Three Variables Together?

Government planning often treats health, sanitation, and energy as separate ministerial portfolios. However, empirical household data proves they are biologically and socially inseparable:

- **The Domestic Environment:** Maternal health (SDG 3.1) is fundamentally compromised by chronic domestic pathogen exposure and indoor air pollution.
- **Pathogen Exposure (SDG 6.2):** Inadequate sanitation causes recurrent enteric infections, stunting nutrient absorption in pregnant women and leading to systemic obstetric complications.
- **Toxic Air & Time Poverty (SDG 7.1):** Reliance on biomass cooking exposes pregnant women to severe particulate pollution (equivalent to smoking 40 cigarettes a day) while increasing physical labor burdens.

2. Interpreting the 2030 Projections

Our time-series validation (trained on 2000–2018, tested on 2019–2023 holdouts) reveals distinct trajectory realities:

Sanitation Plateaus at 76.2% (Drift Model)

While basic latrine construction surged post-2015, upgrading to 'safely managed' systems requires complex sewage treatment. The Drift model projects a 2030 ceiling of 76.2%, missing the 100% target.

MMR Volatility & Massive Uncertainty Bands

MMR's point forecast is 94.3 (missing the <70 target), but its 95% prediction interval spans 7.0 to 181.7. This wide band reflects historical reporting shocks, proving point estimates are misleading for health planning.

Empirical Time-Series Validation & 2030 Projections Summary

Indicator	Selected Model	Validation MAE	2030 Forecast	Lower 95%	Upper 95%	2030 Target	Policy Reality
Maternal Mortality Ratio	ETS	14.89	**94.3**	7.0	181.7	< 70.0	High volatility / Uncertain
Safely Managed Sanitation	Drift	0.06	76.2%	75.4%	77.1%	100.0%	Infrastructure plateau
Clean Cooking Access	ETS	1.93	99.0%	90.9%	107.2%	100.0%	Approaching target

Source: R time-series models trained on 2000–2018 and validated against 2019–2023 holdouts (WHO MMEIG 2025, World Bank JMP, UN SDG Portal).

3. Evidence-Based Government Policy Interventions

Based directly on the forecasting gaps and structural plateaus identified in our time-series data, we recommend four concrete government interventions:

A. Decentralized FSTP Upgrades Municipal Faecal Sludge Grants: To overcome the sanitation plateau (76.2% forecast), shift from subsidizing basic pit latrines to funding decentralized Faecal Sludge Treatment Plants (FSTP) in rural and semi-urban clusters, ensuring safe pathogen containment.	**B. Dynamic LPG Refill Vouchers** Targeted Fuel Cash Transfers: To prevent household fuel stacking and secure 100% clean cooking, insulate low-income families from fossil-fuel price inflation by linking conditional cash subsidies directly to women's bank accounts for recurring cylinder refills.
C. Risk-Band Obstetric Buffers Uncertainty-Aware Health Budgeting: Because MMR forecasting yields a massive 95% upper interval of 181 deaths/100k, state health ministries must abandon rigid point-estimate budgeting and allocate emergency obstetric care reserves based on worst-case risk bands.	**D. Integrated District Convergence Cells** Cross-Sectoral Data Harmonization: Establish district-level convergence cells that merge health (HMIS), sanitation, and energy tracking annually, eliminating reporting lags and aligning local budgeting across silos.

Governance Takeaway for Policy Makers

Sustainable development cannot be achieved by treating individual SDG indicators as isolated checklist items. Our time-series analysis proves that sanitation plateaus and maternal mortality volatility require holistic, cross-sectoral interventions. By aligning sanitation infrastructure, clean energy subsidies, and uncertainty-aware health budgeting, India can secure resilient outcomes heading toward 2030.

Drafted based on India SDG Time-Series Forecasting Project (2000–2024) | Author: Sivachanthini R